

ICAS

PAPER

A



2017 MATHEMATICS

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

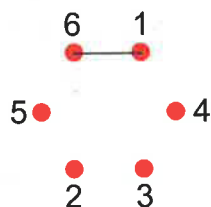
1. This table shows the number of students who ran some laps of a track. There were 14 students altogether.

Number of laps	Number of students
1	
2	
3	
4	

How many students ran **MORE** than 2 laps of the track?

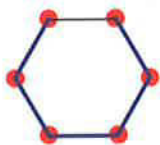
- (A) 3
(B) 4
(C) 6
(D) 8

2. Mia had this dot-to-dot picture.

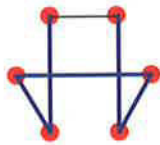


She joined the dots in order from 1 to 6.

Which of these shows what Mia drew?



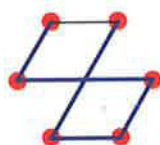
(A)



(B)



(C)



(D)

3. Cai has a vegetable garden.

He grew three rows of tomato plants with four plants in each row.

How many tomato plants did he grow?

- (A) 4
(B) 7
(C) 12
(D) 14

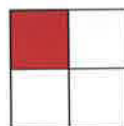
4. Jane filled this large kettle with water.



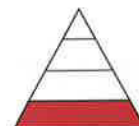
Which is closest to the number of teacups she can fill with this water?

- (A) 3
(B) 15
(C) 60
(D) 100

5. Which of these shapes does **NOT** have one-quarter shaded?



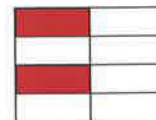
(A)



(B)



(C)



(D)

6. Today is Fadi's birthday.



The symbol  stands for the number of candles on Fadi's birthday cake.

How many candles should be on Fadi's birthday cake in 5 years' time?



+ 5

(A)



- 5

(B)

5 -



(C)

5 ×

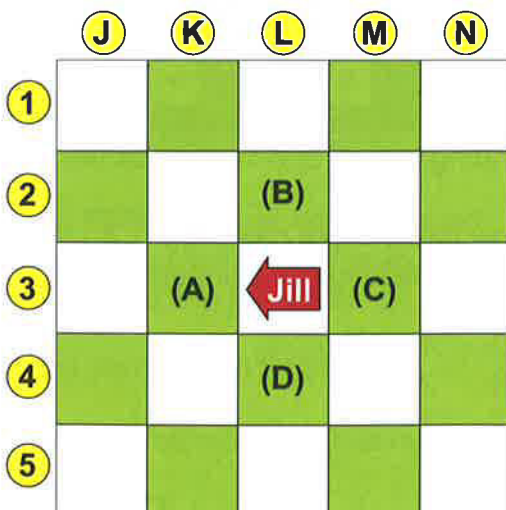


(D)

7. Jill is standing in L3 facing **3**.

She moved one square to her right.

Which square did Jill move to?



8. Mary read the time on the clock.

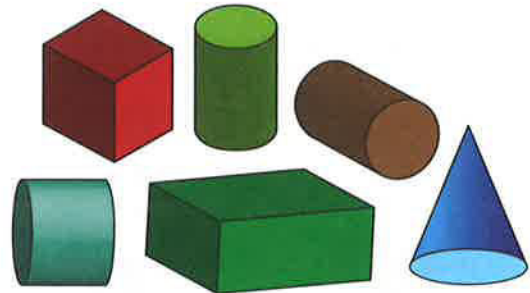


Her class starts in 10 minutes.

What time does Mary's class start?

- (A) 6:08
(B) 10:40
(C) 11:16
(D) 11:40

9. Pat has these solids. The flat faces with no corners are circular.



How many circular faces are there?

- (A) 4
(B) 6
(C) 7
(D) 10

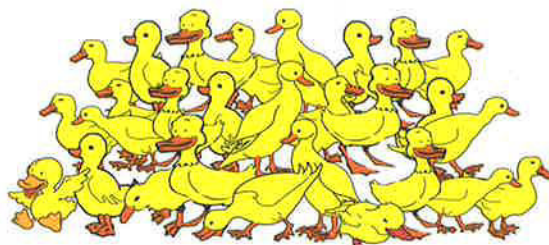
10. Mr Merlino measured the area of his garden.



Which unit should Mr Merlino use?

- (A) millilitres
- (B) kilograms
- (C) centimetres
- (D) square metres

12. Tala had 28 ducklings in a pen. She took them out for a swim in groups of five until there were less than five left in the pen.



How many ducklings were left in the pen?

- (A) 3
- (B) 4
- (C) 5
- (D) 23

11. Rob's plant has two flowers.

Emma's plant has four more flowers than Rob's plant.

Which of these is Emma's plant?



(A)



(B)

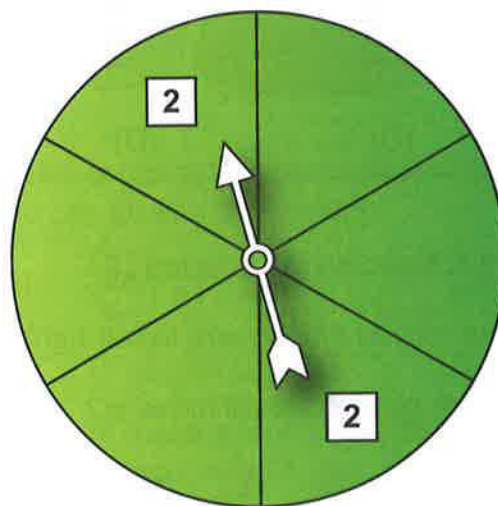


(C)



(D)

13. Sally is making a spinner by placing **2** or **3** in each of the six parts.



She wants the chances of the spinner landing on a 2 or on a 3 to be the same.

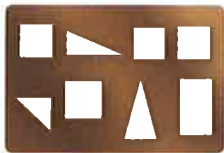
How many more cards with **2** does Sally need to add to her spinner?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

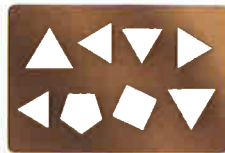
14. Arun rolled out some pastry.

He then cut some triangles and squares out of the pastry. No other shapes were cut out.

Which of these shows Arun's piece of pastry?



(A)



(B)



(C)



(D)

15. Petra used the buttons on her calculator to make a code.



Each number in her code is found by jumping over the 5. For example, 2 becomes 8 in her code.

Petra wrote down 943.

What was her code for this number?

- (A) 1 6 7
- (B) 1 3 7
- (C) 7 9 1
- (D) 7 6 1

16. Mal, Sam and Bob are taking turns on a balance board. Mal is heavier than Sam but lighter than Bob.

Which picture is correct?



(A)



(B)

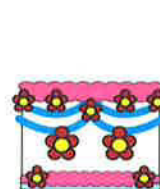


(C)



(D)

17. Which picture will **NOT** look the same after a reflection (flip)?



(A)



(B)



(C)



(D)

18. The diagram shows Tim flying a kite.



Which kite is flying at about the same angle as Tim's?



(A)



(B)



(C)

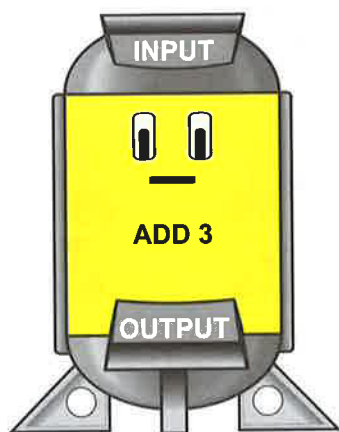


(D)

19. Ms Green made a number machine.

This machine adds 3 to any number put into it.

Carl put only odd numbers through the machine.



What kind of numbers would come out of the machine?

- (A) odd numbers only
- (B) even numbers only
- (C) the number 6 only
- (D) some even and some odd numbers

20. $32 - 8$ is _____ $16 + 9$.

Which of these makes the statement true?

- (A) smaller than
- (B) equal to
- (C) bigger than
- (D) double

21. Mark told only James his secret. James told his friends, Pete and Andy. Pete and Andy each told three different people.

How many people, apart from Mark, know Mark's secret?

- (A) 10
- (B) 9
- (C) 8
- (D) 6

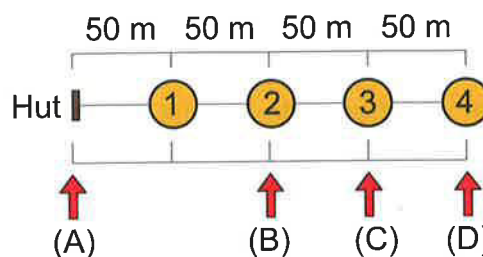
22. A ski lift with cable cars, numbered 1 to 4, has broken down.

Mighty Mitch comes to the rescue.

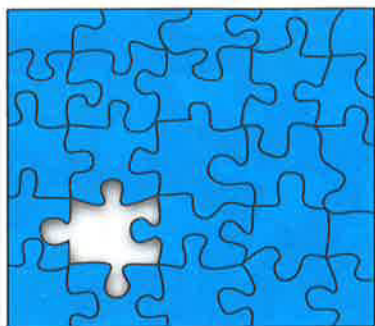


He flies from the hut and brings the closest cable car back to the hut each time.

Where is Mighty Mitch when he has flown a total of 400 metres?



27. Jim has lost a piece of his 'Blue Sky' jigsaw puzzle.



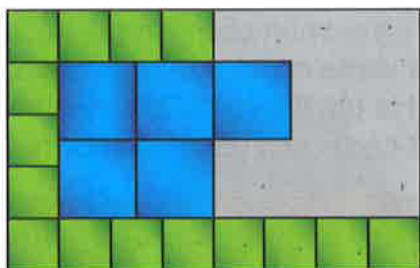
Which is the missing piece?



(A) (B) (C) (D)

28. Gina started tiling a floor using large tiles and small tiles.

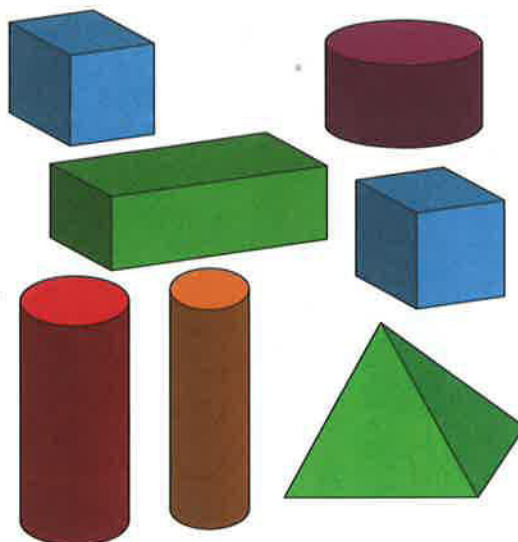
She used the small tiles around the outside and the large tiles in the middle, as shown.



How many tiles will Gina use altogether?

- (A) 28 (B) 30
(C) 34 (D) 40

29. Elke found these seven solids in a box.



She took them out of the box one at a time.

The first solid Elke took out was a pyramid.

Elke then took out a cube.

Which type of solid was impossible for Elke to take out next?

- (A) cube
(B) prism
(C) cylinder
(D) pyramid

30. Betty had these five cards.



Betty chose two cards. One was the largest odd number and the other was the smallest odd number.

What is the difference between these two numbers?

- (A) 50
(B) 54
(C) 63
(D) 70

31. Jack had drum lessons on Monday and Friday afternoons in Term 1.

Term 1

holidays							
	SUN	MON	TUE	WED	THUR	FRI	SAT
January						1	2
	3	4	5	6	7	8	9
	10	11	12	13	14	15	16
	17	18	19	20	21	22	23
	24	25	26	27	28	29	30
	31	term 1 begins					

	SUN	MON	TUE	WED	THUR	FRI	SAT
February		1	2	3	4	5	6
	7	8	9	10	11	12	13
	14	15	16	17	18	19	20
	21	22	23	24	25	26	27
	28	29					

	SUN	MON	TUE	WED	THUR	FRI	SAT
March			1	2	3	4	5
	6	7	8	9	10	11	12
	13	14	15	16	17	18	19
	20	21	22	23	24	25	26
	27	28	29	30	31		

	SUN	MON	TUE	WED	THUR	FRI	SAT
April						1	2
	3	4	5	6	7	8	9
	10	11	12	13	14	15	16
	17	18	19	20	21	22	23
	24	25	26	27	28	29	30

term 2 begins

There were no lessons during holidays.

How many drum lessons did Jack have during Term 1?

- (A) 19
- (B) 20
- (C) 23
- (D) 28

32. Ying and Cai were placing coloured counters on this grid.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Ying placed a green counter on every 7th square. Cai placed a blue counter on every 4th square.

They continued in this way until they were finished.

How many squares have only a green counter on them?

- (A) 4
- (B) 7
- (C) 11
- (D) 14

33. Jane was born on 10 November 2005 while Yan was born on 10 February 2004.

What is their age difference?

- (A) 21 months
- (B) 10 months
- (C) 9 months
- (D) 3 months

34. Jai smudged some of his working.

$$16 - \text{smudge} = 7$$
$$4 \times \text{smudge} = 20$$

He multiplied the two smudged numbers together.

What should Jai's answer be?

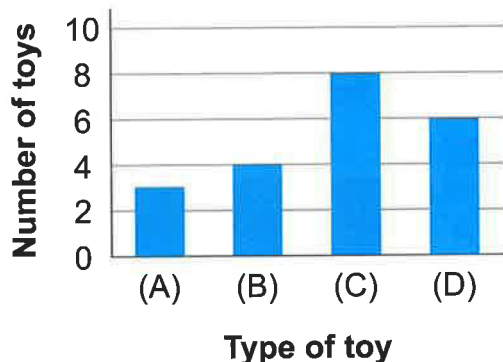
- (A) 14 (B) 40
(C) 45 (D) 55

35. Each student in Anna's class brought a toy to school.

There were twice as many action heroes as there were stuffed toys brought in.

Four students brought in puzzles.

Which column shows the number of action heroes brought to school?



36. Ravi takes 4 paces for every 5 of Isha's paces.

Ali takes 3 paces for every 4 of Isha's paces.

Ravi measures the length of the path with 16 paces.

How many paces will Ali take to measure the same path?

- (A) 12 (B) 15
(C) 17 (D) 20

37. Kevin is solving a puzzle by placing the numbers from 1 to 9 on a grid.

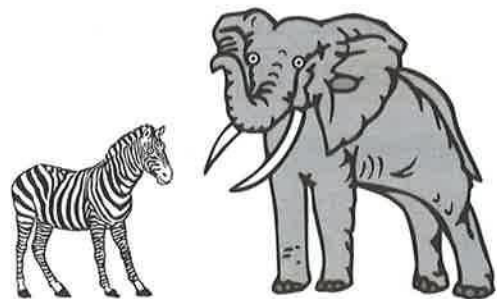
	3	?
1		6

The numbers in each row and in each column must add to 15.

What number must ? be?

- (A) 2
(B) 4
(C) 5
(D) 7

38. The zookeeper measured the heights of an elephant and a zebra.



The sum of their heights was 550 centimetres.

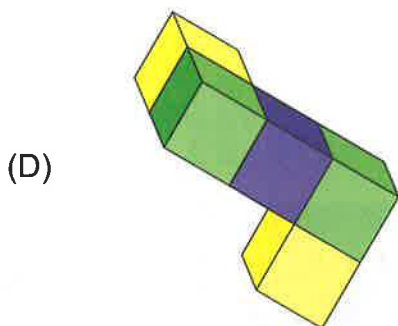
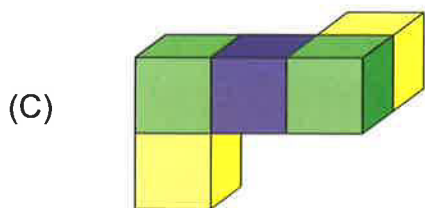
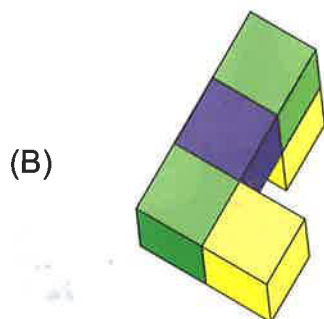
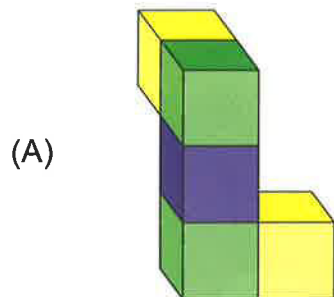
Their height difference was 250 centimetres.

What was the height of the zebra?

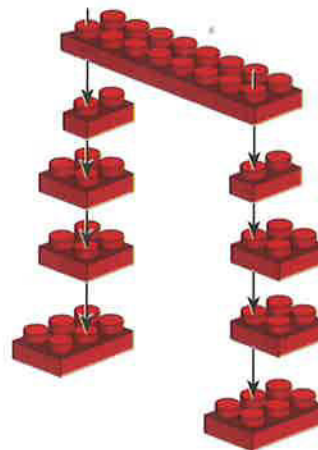
- (A) 150 centimetres
(B) 200 centimetres
(C) 250 centimetres
(D) 300 centimetres

39. Jamie made a solid using cubes. Three of these options are pictures of Jamie's solid and one is not.

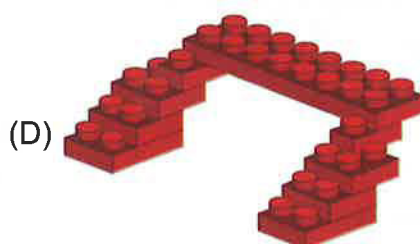
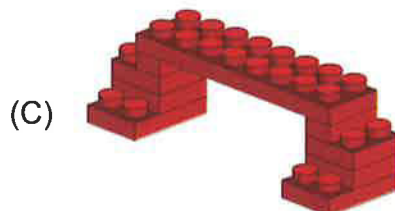
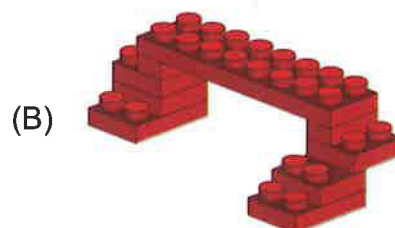
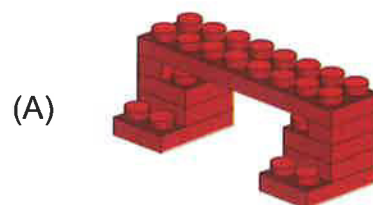
Which one is **NOT**?



40. These are the instructions for building an object out of blocks.



What will the object look like when built correctly?



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Hong Kong	Primary 3
Indian Subcontinent¹	Class 3
Indonesia	Year 4
Malaysia	Standard 3
Middle East²	Class 3
New Zealand/Pacific³	Year 4
Singapore	Primary 2
Southern Africa⁴	Grade 3

¹ Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

² Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Egypt, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria and Jordan.

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AUSTRALIA

ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

	Question content	Area assessed	Question number	Correct answer
Difficult Questions	Use the sum and difference of two heights to find the height of an animal	Number & Arithmetic	38	A
	Count up by 4 and 7 to solve a pattern on a grid	Number & Arithmetic	32	C
	Add interval distances to find a location after a given distance travelled	Measures & Units	22	B
	Identify the column that represents specified information in a graph	Chance & Data	35	D
	Identify the position of a term in a number pattern	Algebra & Patterns	26	C
	Apply relative informal measures of length to solve a problem	Measures & Units	36	B
	Complete a magic square to solve a problem	Number & Arithmetic	37	D
	Identify the piece missing from a jigsaw puzzle	Space & Geometry	27	D
	Follow a given rule to decipher a code	Algebra & Patterns	15	A
	Read a relative position on a grid map	Space & Geometry	7	B
	Apply reasoning to solve a word problem	Number & Arithmetic	21	B
	Identify the solid that is different to the others	Space & Geometry	39	D
	Find the area of a rectangle using informal units	Measures & Units	28	B
	Identify the region that is not one-quarter of a given shape	Number & Arithmetic	5	B
	Identify a word read from the back of a window	Measures & Units	24	D
	Identify the number of circular faces in a group of solids	Space & Geometry	9	C
	Estimate the difference in height of two objects	Measures & Units	23	C
	Find two numbers missing from two equations and calculate their product	Number & Arithmetic	34	C
	Determine the number of months between two given dates	Measures & Units	33	A
	Identify the solid not included in a group	Chance & Data	29	D
Easy Questions	Visualise a completed 3-D object that matches a set of instructions	Space & Geometry	40	C
	Read and interpret a calendar	Measures & Units	31	A
	Determine the rule when adding an odd number to an odd number	Algebra & Patterns	19	B
	Estimate the number of cups that can be filled from a kettle	Measures & Units	4	B
	Complete a number sentence involving an inequality	Number & Arithmetic	20	A
	Identify the cake that is not symmetrical	Space & Geometry	17	D
	Identify the picture with dots connected in the correct order	Space & Geometry	2	D
	Calculate the difference between two odd numbers	Number & Arithmetic	30	B
	Create a spinner with two events having the same chance of success	Chance & Data	13	A
	Identify the object that makes the same angle with the ground	Space & Geometry	18	A
	Identify the diagram that represents given information	Measures & Units	16	B
	Calculate the remainder after repeatedly subtracting five	Number & Arithmetic	12	A
	Interpret a pictograph where one symbol represents one object	Chance & Data	1	C
	Read an analogue clock and convert to digital time	Measures & Units	8	D
	Identify the object least likely to be chosen	Chance & Data	25	A
	Choose the appropriate unit of measurement for area	Measures & Units	10	D
	Create an expression to represent a given situation	Algebra & Patterns	6	A
	Identify the pattern with only squares and triangles	Space & Geometry	14	C
	Solve a simple addition problem using a range of strategies	Number & Arithmetic	11	D
	Recognise multiplication in context	Number & Arithmetic	3	C